Page 1: Title Page

Project Title: Banking Analytics Dashboard: A Data-Driven Approach to Financial Intelligence

Document Type: Technical Project Report

Tools Used: Microsoft SQL Server, Power BI, DAX, Power Query

1.3 Executive Summary

The **Banking Analytics Dashboard** project represents a comprehensive Business Intelligence (BI) solution developed to convert raw financial and transactional data into meaningful, decision-driven insights. Financial institutions generate massive volumes of data daily, yet deriving strategic value from this information requires structured data models, analytical logic, and intuitive visualization.

This project leverages a relational database architecture built in SQL Server alongside dynamic reporting capabilities in Power BI. The solution enables stakeholders to monitor liquidity positions, evaluate customer behavior, and uncover transaction patterns that influence operational and strategic decisions. By integrating data engineering, analytics, and visualization, the dashboard demonstrates how modern BI systems support data-driven banking operations.

1.2 Introduction

This project addresses that challenge by simulating a real-world banking analytics scenario. Instead of merely storing data, the focus is placed on transforming and analyzing it to reveal operational trends, behavioral patterns, and performance indicators. The initiative demonstrates how analytical frameworks and visualization tools can bridge the gap between raw data and executive decision-making.

1.3 Project Scope

The scope of the Banking Analytics Dashboard project spans the **end-to-end Business Intelligence lifecycle**, covering the technical, analytical, and presentation layers of a modern BI solution:

- **Data Engineering**
  The project begins with designing a structured relational schema within SQL Server to ensure data integrity, consistency, and scalability. SQL-based transformations, cleansing processes, and calculated fields are implemented to prepare raw datasets for analytical use. This stage emphasizes normalization, relationship management, and performance optimization.

- **Data Analysis**
  Analytical logic is applied to derive key financial and operational metrics. Important KPIs such as **Average Daily Balance**, **Transaction Velocity**, account activity levels, and liquidity indicators are calculated to assess banking performance. This phase focuses on uncovering patterns, identifying anomalies, and generating insights that reflect customer and transactional behavior.

- **Data Visualization**
  The final layer involves constructing intuitive and interactive Power BI reports that translate complex data into easily interpretable visuals. Dashboards are designed with executive usability in mind, enabling users to explore trends, compare metrics, and make informed decisions through dynamic filtering and drill-down capabilities.

- **Business Value & Decision Support**
  Beyond technical implementation, the project emphasizes how BI systems enhance decision-making by providing visibility into financial operations, risk factors, and customer dynamics. The dashboard acts as a strategic tool rather than a static report.

Page 3: Data Architecture & Modeling

2.1 The Relational Schema

The Banking Analytics Dashboard is built using a **Star Schema** design. This structure organizes data in a way that makes analysis faster and easier.

In a star schema, one main table (the fact table) stores measurable data, while surrounding tables (dimension tables) store descriptive information. This approach improves query performance, simplifies relationships, and allows Power BI to generate reports efficiently.

By using this design, the dashboard can quickly analyze large volumes of banking transactions without unnecessary complexity.

2.2 Entity Descriptions

The data model includes one central fact table and multiple dimension tables:

- **Fact_Transactions**
  This is the main table containing transaction data such as transaction amount, date, and type. It represents the core financial activities used for analysis.
- **Dim_Customers**
  Stores customer-related details like age, gender, and location. This table helps analyze customer behavior and demographics.
- **Dim_Accounts**
  Contains account information such as account type (Savings or Checking) and account status (Active or Dormant). This enables account-level analysis.
- **Dim_Currency**
  Provides currency information and supports multi-currency analysis and conversions, ensuring accurate financial reporting.

2.3 SQL Data Transformation

Several SQL operations were performed to prepare the data for analysis:

1. **Deduplication**
   Duplicate transaction records were removed to prevent incorrect calculations and reporting errors.
2. **Aggregations & Views**
   SQL views were created to summarize data, such as monthly balances and transaction totals. This improves performance and speeds up Power BI reports.
3. **Data Quality Checks**
   Validation rules ensured accurate values, valid dates, and correct relationships between tables.

Page 4: Analytical Methodologies

3.1 Key Performance Indicators (KPIs)

The dashboard monitors several important metrics that reflect financial health and customer behavior:

- **Total Liquidity**
  Represents the combined balances of all accounts within the institution. This metric helps evaluate the bank's available funds and overall financial position.
- **Churn Risk Indicator**
  Identifies accounts with no transaction activity for more than 90 days. Such inactivity may signal potential customer attrition or dormant accounts.
- **Credit-to-Debit Ratio**
  Compares incoming funds (credits) to outgoing funds (debits). This measure provides insight into cash flow patterns and customer spending behavior.

These KPIs provide decision-makers with a quick snapshot of institutional performance and risk factors.

3.2 DAX Measures (Power BI)

Advanced calculations were implemented using **DAX (Data Analysis Expressions)** to support dynamic analysis:

- **Total Balance**
  Calculates the sum of all account balances to provide an aggregated financial view.
- **Year-over-Year (YOY) Growth**
  Compares current-year transaction volumes with the previous year, helping identify growth trends or seasonal patterns.
- **Customer Segmentation**
  Classifies customers into categories such as *High Value*, *Mid-Tier*, and *Retail* based on balance thresholds. This enables targeted analysis and strategic planning.

These measures allow the dashboard to respond dynamically to filters and user interactions.

3.3 Transaction Trend Analysis

Time-based analysis was applied to evaluate transaction patterns over different periods. This approach helps identify peak activity times, seasonal fluctuations, and behavioral trends.

Understanding these trends allows banks to better manage liquidity, forecast demand, and anticipate system usage requirements.

Page 5: Dashboard Insights & Features

4.1 Visual Hierarchy

The dashboard is structured into three main analytical views:

1. **Executive Overview**
   Displays high-level metrics such as Total Assets and Total Customers, offering a quick performance summary for decision-makers.

2. **Demographic Pane**
   Uses maps and charts to visualize customer distribution, regional activity, and sources of financial inflows.

3. **Operational Pane**
   Provides detailed breakdowns of transactions, including Debit vs. Credit activity and currency usage.

This layout ensures clarity, usability, and efficient navigation across different analytical needs.

4.2 Interactive Features

Several interactive elements enhance the user experience:

- **Slicers**
  Allow users to filter reports by Date Range, Account Type, or Region, enabling flexible exploration.

- **Drill-Through**
  Enables deeper analysis by allowing users to click on specific data points (such as a customer) to view detailed records.

- **Tooltips**
  Display additional information when hovering over visuals, improving data interpretation without cluttering the interface.

These features make the dashboard more engaging and analytically powerful.

4.3 Key Findings

The dashboard reveals several meaningful insights:

- **High-Value Concentration**
  A small percentage of customers typically holds a large share of total balances, highlighting revenue and risk concentration.

- **Currency Usage Patterns**
  Certain foreign currencies generate higher transaction volumes and may contribute significantly to exchange-related revenue.

Such findings support strategic planning and financial decision-making.

Page 6: Business Impact & Conclusion

5.1 Business Impact

The Banking Analytics Dashboard delivers value across multiple business functions:

- **Risk Management**
  Helps detect unusual transaction behavior that may indicate fraud or operational anomalies.
- **Marketing Strategy**
  Supports customer segmentation for personalized product offerings and targeted campaigns.
- **Operational Efficiency**
  Automates complex reporting processes, reducing manual effort and improving decision speed.

5.2 Learning Outcomes & Technical Growth

The development of this project strengthened several technical and analytical skills:

- Advanced SQL techniques, including joins and analytical functions
- Effective dashboard design and data storytelling principles
- Translating business questions into analytical solutions

5.3 Final Conclusion

The Banking Analytics Dashboard demonstrates how structured data modeling, analytical logic, and visualization tools can transform raw financial data into actionable insights.

Rather than being just a reporting interface, the dashboard represents a practical example of how Business Intelligence supports informed decision-making and digital transformation in financial services.